

Diagonal solutions do not give the proposed VMVT coefficient

Disproof of Conjecture 00000004157

gaochengzhecpu

Abstract

For $k = 1, s = 2$, inside the stated range $s \leq k(k+1)$, the exact Vinogradov mean value is $(2N^3 + N)/3$, whereas the diagonal count is $2N^2 - N$. Thus the true cubic coefficient is $2/3$, and the diagonal contribution on that same scale is zero. The diagonal count's own leading coefficient is $2! = 2$, which also differs from $2/3$. As a supplementary degree-two obstruction, an elementary counting argument gives $J_{6,2}(N) \geq N^9/49$, while there are at most $6!N^6$ diagonal solutions. The Lean project proves these facts for the actual finite solution sets.

Statement and conventions

The source defines the optimal constant as the main-term coefficient of $J_{s,k}(X)$. It asserts that this coefficient always equals the number of diagonal solutions, expressed by factorials, with validity exactly for $s \leq k(k+1)$. It imposes no restriction excluding $k = 1$.

For an integer $N \geq 1$, use the standard unweighted count

$$J_{s,k}(N) = \# \left\{ (\mathbf{x}, \mathbf{y}) \in \{1, \dots, N\}^{2s} : \sum_{i=1}^s x_i^j = \sum_{i=1}^s y_i^j \quad (1 \leq j \leq k) \right\}.$$

By a diagonal solution we mean an ordered pair for which $\mathbf{y}$ is a permutation of $\mathbf{x}$; each pair is counted once, even when repeated coordinates allow several permutations. Write its count as $D_s(N)$. These are the usual solution counts for the Vinogradov mean value. Disagreement along the integer values of X suffices to refute an asserted general asymptotic.

An exact counterexample to the coefficient claim

Theorem 1. *For every integer $N \geq 1$,*

$$J_{2,1}(N) = \frac{2N^3 + N}{3}, \quad D_2(N) = 2N^2 - N.$$

Consequently

$$\lim_{N \rightarrow \infty} \frac{J_{2,1}(N)}{N^3} = \frac{2}{3}, \quad \lim_{N \rightarrow \infty} \frac{D_2(N)}{N^3} = 0, \quad \lim_{N \rightarrow \infty} \frac{D_2(N)}{N^2} = 2.$$

Proof. The only equation is $a + b = c + d$. Translating all four coordinates by -1 changes the domain to $\{0, \dots, N-1\}$ without changing this equation. Classify a solution by $\delta = a - c = d - b$.

When $\delta = 0$, there are N^2 solutions. If $1 \leq \delta \leq N-1$, choose $c, b \in \{0, \dots, N-1-\delta\}$ freely; the remaining coordinates are uniquely $a = c + \delta$ and $d = b + \delta$. Thus there are $(N - \delta)^2$ such solutions. Interchanging the left and right pairs gives the same count for $-\delta$. Hence

$$J_{2,1}(N) = N^2 + 2 \sum_{m=0}^{N-1} m^2 = N^2 + \frac{N(N-1)(2N-1)}{3} = \frac{2N^3 + N}{3}.$$

For the diagonal count the two possible right pairs are (a, b) and (b, a) . Each choice gives N^2 ordered solutions, and the two families intersect exactly when $a = b$, in N solutions. Therefore $D_2(N) = 2N^2 - N$. Dividing these exact formulas by the displayed powers of N proves the limits. $\square$

Here $s = 2 = k(k + 1)$ for $k = 1$, so the example belongs to the claimed range. On the actual main-term scale N^3 , the coefficient of $J_{2,1}$ is $2/3$ while the diagonal contribution is zero. If the wording is instead read as asserting equality with the factorial coefficient of the diagonal count itself, that coefficient is $2! = 2$, again unequal to $2/3$. Thus the counterexample directly concerns the coefficient, not only the existence of some non-diagonal solutions.

A supplementary obstruction with degree two

There is also a failure of diagonal main-term dominance at $k = 2, s = 6$, again with $s = k(k + 1)$. Map each $\mathbf{x} \in \{1, \dots, N\}^6$ to its genuine moment pair

$$F(\mathbf{x}) = \left(\sum_{i=1}^6 x_i, \sum_{i=1}^6 x_i^2 \right).$$

Both entries are nonnegative integers, at most $6N$ and $6N^2$, respectively. Thus they lie in a box with

$$B_N = (6N + 1)(6N^2 + 1) \leq 49N^3$$

elements. If r_b is the number of tuples in the fiber over b , then

$$\sum_b r_b = N^6, \quad J_{6,2}(N) = \sum_b r_b^2.$$

Cauchy–Schwarz gives

$$N^{12} \leq B_N J_{6,2}(N) \leq 49N^3 J_{6,2}(N), \quad J_{6,2}(N) \geq \frac{N^9}{49}.$$

Choosing the left tuple and one of the $6!$ permutations covers every diagonal pair, possibly with repetitions. Hence $D_6(N) \leq 720N^6$. It follows that $J_{6,2}(N)$ is not $O(N^6)$, and in particular cannot be asymptotic to $D_6(N)$. This is a supplementary refutation of diagonal dominance within the claimed range; it does not purport to compute the exact degree-two asymptotic coefficient. The preceding degree-one example already supplies an exact coefficient counterexample to the source as written.

Formal correspondence and verification

The Lean project works with actual finite types and their cardinalities. For $k = 1, s = 2$, `LinearSolutions` is the subtype of four entries in `Fin N` satisfying $a + b = c + d$. The theorem `linear_equation_iff_standard` proves that adding one to all entries gives the standard equation on $\{1, \dots, N\}$. The map `encodeLinear` implements the three difference cases in the proof; its injectivity and surjectivity are proved before its cardinality is used. The diagonal set is the union of the identity and transposition families, with its intersection counted exactly.

The theorems `J21_exact` and `D21_exact` establish the two polynomial formulas. The actual real limits, including the zero diagonal contribution on the cubic scale, are proved using `Filter.Tendsto`. They are combined with the range check in `explicit_coefficient_counterexample`.

For $k = 2, s = 6$, `momentMap` sends actual six-tuples to the two integer power sums. The theorem `momentMap_eq_iff` identifies collisions with the two Vinogradov equations. An explicit

equivalence with a disjoint union of pairs of fibers proves the collision formula; finite Cauchy–Schwarz gives the lower bound. The diagonal type consists of actual permuted tuple pairs, and these are proved to satisfy both equations. The project then proves the $720N^6$ bound and negates asymptotic equivalence with the full solution count.

The project is pinned to Lean 4.19.0 and a fixed Mathlib commit. No custom axioms, proof placeholders, or numerical oracles are used. Printed theorem dependencies are limited to the standard `propext`, `Classical.choice`, and `Quot.sound`. No external theorem about the VMVT asymptotic is assumed.

Source

The Last Math Competition, Conjecture 00000004157. The submission preserves the exact bilingual source and records its commit and SHA-256 separately.